

Establishing *wMelPop* *Wolbachia* Infection among Wild *Aedes aegypti* Females by Optimal Control Approach

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Abstract: *Wolbachia* is a maternally transmitted bacterial symbiont which is known to reduce the vector competence of mosquitoes and other arthropod species. Therefore, *Wolbachia*-based biocontrol is regarded as a practicable method for prevention and control of dengue and other arboviral infections. In particular, a deliberate infection of *Aedes aegypti* females with *wMelPop* *Wolbachia* strain makes them almost incapable of transmitting dengue and other arboviruses. In this paper, we present and thoroughly analyze a population dynamics model of interaction between wild *Aedes aegypti* female mosquitoes and those infected with *wMelPop* *Wolbachia* strain, which compete for the same vital resources (food, breeding sites, etc.) and share the same locality. Using this model, we demonstrate that the final outcome of the competition essentially depends on the frequency of *Wolbachia* infection. Further, we apply the optimal control approach and design the control intervention programs based on periodic releases of *Wolbachia*-carrying females for establishing *wMelPop* *Wolbachia* infection in the target locality.

Keywords: population dynamics, biological control, *wMelPop* *Wolbachia* strain, *Aedes aegypti*, optimal control

1 Introduction

Aedes aegypti is an invasive mosquito species which is widely spread in urban and semi-urban areas of all tropical and sub-tropical regions worldwide [7]. The presence and abundance of this mosquito species is strongly correlated with persistence and proliferation of different arboviral infections, such as yellow fever, dengue, chikungunya, and zika. The virus is transmitted by *Aedes aegypti* females during their blood meals taken on human individuals. Therefore, the infection spread can be controlled either by suppressing the mosquito population or by reducing the virus transmissibility.

Effectively, the transmissibility of different arboviral infections can be drastically reduced when their transmitters (i.e., *Aedes aegypti* females) carry an intracellular bacterial symbiont of *Wolbachia*¹ [10,11,26,40]. This biological agent is maternally inherited by the offspring at eggs stage and induces a particular reproductive phenotype of *cytoplasmic incompatibility*

(CI). According to [7, p.450], CI-phenotype causes inviability of offspring originated from matings between *Wolbachia*-free females and *Wolbachia*-carrying males. On the other hand, the offspring of *Wolbachia*-infected female that originates from her mating with either infected or uninfected male will be infected with *Wolbachia* pathogen. The latter facilitates the spread of *Wolbachia* infection in wild insect populations by conferring a certain reproductive advantage to *Wolbachia*-carrying females in producing viable offspring.

There are three major *Wolbachia* strains (*wMelPop*, *wMel*, and *wAlb*) that are currently being tested for control and prevention of arboviral infections [16]. These strains have different potentials which are measured in terms of their fitness costs versus their abilities to reduce the vector competence and to block the virus transmission. In particular, *wMelPop* *Wolbachia* strain has higher fitness cost since it reduces the female fecundity, the viability of eggs, and the lifespan of infected mosquitoes to a greater extent than other two strains [15, 25,32,34,35,40]. Nonetheless, *wMelPop* strain has more benefits for prevention and control of arboviral infections

¹ A comprehensive description of *Wolbachia* symbiont can be consulted in [7, p.437].

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since it confers stronger inhibition of virus replication in *Aedes aegypti* mosquitoes and reduces the proportion of older mosquitoes, which are responsible for the majority of disease transmission [11,26,41].

The spread of *Wolbachia* infection among wild population of mosquitoes can be achieved by large-scale field releases of *Wolbachia*-carriers in some target locality. Such *Wolbachia*-carriers are “cultivated” in laboratory condition via artificial breeding of mosquito eggs, which are deliberately infected (or *transinfected*) with *Wolbachia* pathogen taken from other insect species [24]. Different type of release programs have been proposed and tested by various scholars [11,14,16,24,28,32,35,40,42]. The majority of the release programs have a strongly marked inundative character, which is appropriate when the fitness of *Wolbachia*-carrying invaders and *Wolbachia*-free local residents is comparable while CI reproductive phenotype acts in favor of *Wolbachia*-carriers.

However, the reduced fitness of mosquitoes infected with wMelPop *Wolbachia* strain makes their invasion, persistence, and future population replacement rather challenging and requires to design a new type of release programs. Several cases reporting the failure of establishing wMelPop *Wolbachia* strain among wild populations of *Aedes aegypti* mosquitoes by traditional release programs have been recently communicated in literature [28,42].

This paper intends to contribute in this strand of research by designing a new type of release program for replacement of wild mosquitoes by those carrying the wMelPop strain of *Wolbachia*. For this purpose, we propose a stylized model (see Section 2) that describes the population dynamics of *Aedes aegypti* female mosquitoes, which act is the major transmitters of different arboviral infections². This model accounts for the density-dependence and competition between uninfected and *Wolbachia*-infected females for the same resources (such as food, breeding sites, etc.) available to both subpopulations (or species). Additionally, our model comprises the critical depensation term in the recruitment part of wild mosquito dynamics. This term is introduced in order to express mathematically the fact that, at high frequencies of *Wolbachia* infection, wild females (despite possessing a better fitness!) have fewer possibilities to produce viable offspring than their *Wolbachia*-infected coevals. This fact comes from scientific evidence and has been reflected in other models describing *Wolbachia* invasion [3,5,6,35,39]. Furthermore, our model allows to estimate the threshold in the population size of wild female mosquitoes below which an average uninfected female produces less viable offspring than an average *Wolbachia*-infected female. When the population of wild females drops below this threshold at sufficiently high frequency of *Wolbachia* infection, the population of wild

mosquitos should be driven towards extinction and be ultimately replaced by the *Wolbachia*-carriers. In other words, *Wolbachia* invades the mosquito population and the population replacement occurs.

In Section 3 we formulate the problem of optimal control where the control action consists in releasing a certain number of *Wolbachia*-infected female mosquitoes into the wild *Aedes aegypti* populations at each day t and this number may vary from one day to another within a given range. The final goal is to replace the wild mosquito population by the *Wolbachia*-carriers while minimizing the overall time of control intervention together with the total costs related to implementation of the control policy.

Section 4 is focused on the numerical solution of the optimal control problem under eight different scenarios divided in two groups. Each group corresponds to a certain assignment of the decision-making priorities, while four scenarios within each group deal with different level of maximum daily release capacities. All numerical experiment have been carried out using the entomological parameters of wMelPop strain of *Wolbachia*, which is regarded as the best blocker of dengue and other arboviruses [11,26,41] but is possesses an extremely high fitness cost [15,25,32,34,35,40] what makes it challenging to establish in wild mosquito populations. The outcome of each scenario renders the optimal (minimum) time of control intervention and the optimal daily release program under which the population of wild mosquitoes is driven towards extinction thus allowing for wMelPop *Wolbachia* invasion to occur. All eight scenarios are also analyzed from the standpoint of their potential efficiency or cost-effectiveness, and the most appropriate scenario is identified.

Finally, Section 5 presents the conclusions, final comments, and perspectives of future research.

2 Population dynamics of *Aedes aegypti* females

2.1 Model formulation

Let $M(t)$ be the total number of *Aedes aegypti* females present in the target locality at the day t . According to [23], the density-dependent mosquito birth (or recruitment) rate can be expressed as

$$h(M) = \Psi - \frac{r}{K}M, \quad (1)$$

where $\Psi > 0$ denotes the natural birth rate of adult female mosquitoes in absence of the density dependence, $r = \Psi - \delta > 0$ stands for intrinsic growth rate of female mosquitoes in absence of the density dependence, $\delta > 0$ is the natural mortality rate of adult female mosquitoes in absence of the density dependence, and $K > 0$ denotes the carrying capacity of the environment (i.e. the maximal number of female mosquitoes supported by the environment).

² It is worthwhile to note that male mosquitoes do not bite people and neither transmit arboviral infections.

In the above equation, the negative term refers to the density dependence caused by the limited availability of breeding sites and competition at larval stage [14,20]. Using the density-dependent growth rate (1), the population dynamics of *Aedes aegypti* females can be described by logistic equation

$$\begin{aligned}\frac{dM}{dt} &= h(M)M - \delta M = \left(\Psi - \frac{r}{K}M\right)M - \delta M \\ &= r\left(1 - \frac{M}{K}\right)M\end{aligned}\quad (2)$$

which is in line with sex-structured model for *Aedes aegypti* population dynamics [6] according to which female mosquitoes have “almost” logistic growth. Model (2) requires to assume the presence of male mosquitoes in the environment and this population compartment is not explicitly present in the model. For the sake of simplicity, let us suppose that at each moment $t \geq 0$, the number of living male mosquitoes is proportional to $M(t)$ with proportionality constant less than 1. This assumption is quite logical since the mosquito sex ratio at birth is about 1 : 1 while males have a shorter lifespan than females [22].

Now let us suppose that the population of *Aedes aegypti* females consists of two groups, that is, $M(t) = W(t) + F(t)$, where $W(t)$ denotes the share of *Wolbachia*-carrying females, while $F(t)$ stands for the share of wild (or *Wolbachia*-free) females. Both groups should compete for the same breeding sites and food resources available in the target locality, as well as for mating opportunities.

It is fair to assume that each mosquito group ($F(t)$ and $W(t)$) grows logistically in the absence of another. Therefore, when both groups are present in the same locality, their competition for breeding sites and food resources can be modeled by applying the *principle of competitive exclusion* (see, e.g., [4] or other similar textbooks) according to which only one of two competing species should ultimately survive. The final competition outcome depends on different biological characteristics of the competing species which are referred to as “relative fitness” of each group. In biological terminology, fitness is often defined as a potential propensity (or probability) of the biological species for survival to adult stage and successful reproduction. According to [4], relative fitness of species in competition may be density-dependent or frequency-dependent; the latter implies that the competition outcome also depends on the (relative) size of each population group present in the same locality.

Many scholars point out that wild mosquitoes have better individual fitness than *Wolbachia*-carriers, especially those infected with *wMelPop*, since this particular strain of *Wolbachia* reduces the female fecundity, viability of eggs, and the lifespan of infected mosquitoes [15,25,32,34,35]. On the other hand, *Wolbachia* pathogen is transmitted maternally from an infected female mosquito to her offspring and the presence of this pathogen in mosquito organism grants

the reproductive advantage to *Wolbachia*-carrying female through the mechanism of cytoplasmic incompatibility (CI). CI-reproductive phenotype guarantees the spread of *Wolbachia* infection to offspring engendered by mating of *Wolbachia*-carrying females with either infected or uninfected males, while ensuring the absence of viable offspring as a result of mating between uninfected females and *Wolbachia*-carrying males. Thus, *Wolbachia* infection has the potentials to spread among wild mosquitoes despite their better individual fitness.

The above arguments are reflected in different mathematical models describing *Wolbachia* invasion in wild mosquito populations (see, e.g., [3,5,6,35,39] among others) and all these models have the common feature that consists in the bistable nature of *Wolbachia* dynamics and the existence of certain threshold in the infection frequency³ above which the invasion and stabilization of *Wolbachia* can be achieved. The latter has a simple explanation since wild males and females are scarce at high frequencies of *Wolbachia* infection, while *Wolbachia*-carriers (both males and females) are abundant. Therefore,

- an average wild female has more chances to encounter with *Wolbachia*-carrying male (rather than a wild one) and to produce inviable eggs;
- an average wild male has more chances to encounter with *Wolbachia*-carrying female (rather than a wild one) and to produce *Wolbachia*-infected and viable offspring.

Taking as the basis the competition model described in [4, p.67] and adding the property of frequency-dependent bistability induced by CI reproductive phenotype, we propose the following model for population dynamics and interaction of uninfected and *Wolbachia*-infected *Aedes aegypti* females:

$$\frac{dF}{dt} = \left[\Psi_f - \frac{r_f}{K_f}(F+W)\right]F\left(\frac{F}{K_0} - 1\right) - \delta_f F, \quad (3a)$$

$$\frac{dW}{dt} = \left[\Psi_w - \frac{r_w}{K_w}(F+W)\right]W - \delta_w W. \quad (3b)$$

Here, the meanings of parameters Ψ, r, δ with subindexes (f, w) are the same as in (2), where $\square_f$ refers to *Wolbachia*-free females and $\square_w$ refers to *Wolbachia*-carrying females. However, the meanings of K_f, K_0 and K_w deserve further explanations. First, equation (3a) has “almost” logistic behavior in the absence of *Wolbachia*-carriers (that is, for $W(t) = 0$). In this case, the model can be still considered realistic only if we set $K_0 = 0$. However, our interest consist in modeling the interaction (competition) of two mosquito species; therefore, the presence of *Wolbachia*-carrying female in the dynamical system (3) is compulsory and

³ By infection frequency we understand the share of infected individuals with respect to total mosquito population, that is, $W/(F+W)$.

cannot be ignored. The terms inside square brackets in (3) refer exactly to competition between wild and *Wolbachia*-carrying females for the same resources, which plainly agrees with traditional competition models (see, e.g., [4] or similar textbooks).

Second, it should be recalled that the *wMelPop* *Wolbachia* strain considerably reduces the individual fitness of *Wolbachia*-carrying mosquitoes and the latter is reflected by the following relationship between mosquito-related parameters:

$$\Psi_w < \Psi_f, \quad r_w < r_f, \quad \delta_w > \delta_f. \quad (4)$$

Condition (4) states that uninfected females are more fertile and live longer than their infected coevals. Additionally, there is evidence that *Wolbachia* reduces the insect host density at steady state [9] and, therefore, we can assume that $K_w < K_f$.

Third, equation (3a) includes the critical depensation term $\left(\frac{F}{K_0} - 1\right)$ in the recruitment part, which becomes negative when $F(t) < K_0$ and stays positive for $F(t) > K_0$. Depensation is often refers to so-called *Allee effect* (see, e.g. [4] or similar textbooks) that basically asserts that an average individual fitness tends to declines at low population densities. In the context of our competition model, the depensation term is introduced in the recruitment part of (3a) in order to express the effect that CI reproductive phenotype may have on the final outcome of competition. There is evidence that *Wolbachia* invasion of wild insect population is frequency-dependent, and this fact has been clearly reflected in several frequency-based models [35,39]. These models point out that *Wolbachia* spreads and persists only when an infected female produces, in average, more offspring than an uninfected female.

Thus, at low infection frequencies ($W(t) \ll F(t)$), *Wolbachia*-infected females have no problems with successful mating. They are able to produce viable and *Wolbachia*-infected offspring after mating with either wild or *Wolbachia*-carrying males. However, *Wolbachia*-carriers cannot sustain competition with wild mosquitoes due to their reduced fitness (cf. conditions (4)) and will be eventually driven towards extinction.

On the other hand, at high infection frequencies ($W(t) \gg F(t)$), wild females have more chances to mate with *Wolbachia*-infected than uninfected males (because they are even scarcer than $F(t)$) and the resulting offspring will be inviable. Thus, each new generation of mosquitoes should have a greater share of *Wolbachia*-carriers than the previous one, and wild mosquitoes should eventually become extinct. Frequency-based models reveal the bistable nature of *Wolbachia* dynamics and are characterized by the existence of a certain “infection frequency threshold” above which the invasion and stabilization of *Wolbachia* can be achieved.

However, our model (3) is not frequency-based; therefore, the underlying “infection frequency threshold” can be expressed in this competition model by introducing the concept of “minimum viable population size” (MVPS) of uninfected females, which is always present (explicitly or implicitly) in population dynamics models comprising the Allee effect. The underlying MVPS threshold, which we denote further on as K_b , can be explicitly defined by the model’s parameters K_f, K_0, Ψ_f , and δ_f (see Subsection 2.2, formula (6)) and the presence of this threshold is modeled by the depensation term $\left(\frac{F}{K_0} - 1\right)$ in the recruitment part of the equation (3a). Therefore, only $F(t)$ exhibit the Allee effect at high infection frequency despite their better individual fitness and become unable to sustain competition with $W(t)$.

The MVPS threshold K_b can be estimated by K_0 (in effect, $K_b \geq K_0$, cf. formula (6)). According to equation (3a), the instantaneous growth of wild females should be negative ($\frac{dF}{dt} < 0$) when $F(f) < K_b$ and be positive ($\frac{dF}{dt} > 0$) when $F(f) > K_b$. Therefore, parameter K_0 is related to the population size of $F(t)$ when the infection frequency $\left(p(t) = \frac{W(t)}{F(t) + W(t)}\right)$ has reached its steady state $0 < \hat{p} < 1$ within the limits of the carrying capacity of the environment. For *wMelPop* *Wolbachia* infection in wild *Aedes aegypti* the value of $\hat{p}$ has been estimated in the range 70 – 90% by different authors [11,15,28,32,35,41,42].

It should also be noted that $K_f > 0$ in (3a) is a parameter related to the carrying capacity of uninfected mosquitoes and this capacity itself, which we denote further on as K_* , can be defined explicitly in terms of other parameters referring to wild mosquitoes, that is, K_f, K_0, Ψ_f , and δ_f (see Subsection 2.2, formula (7)).

Remark 1 The use an explicit frequency dependence in the recruitment term of (3a) may seem more appropriate than the introduction of the depensation term. However, we have to keep a viable balance between the biological realism of the model and its simplicity in order to perform an optimal decision making that seeks to establish the *Wolbachia* infection by means of continuous releases of *Wolbachia*-carriers in the target locality. The major drawback of our stylized model (3) is its requirement to account for the presence of both mosquito species in the environment. On the other hand, the model’s highlights consist in its straightforwardness, its bistable nature and its ability to model the right outcome of the competition.

Finally, it is worthwhile to point out that the competition model (3) presented in this paper differs considerably from the model proposed in [5], where the depensation was supposed to be stronger and affected both the recruitment and mortality of wild females. However, under current setting of the model (3), the depensation (or

Allee effect) only affects their reproduction success due the lesser chances of finding an infected partner at high frequencies of *Wolbachia* infection.

2.2 Stability appraisal

Dynamical system (3) is competitive, and competition means that an increase in the size of either population tends to decrease the growth rate of the other population. Let $f(F, W)$ and $g(F, W)$ denote the growth rates of wild females F and *Wolbachia*-infected females W , respectively. These growth rates are defined by the right-hand sides of the system (3), that is,

$$f(F, W) = \left[\Psi_f - \frac{r_f}{K_f}(F + W) \right] F \left(\frac{F}{K_0} - 1 \right) - \delta_f F, \quad (5a)$$

$$g(F, W) = \left[\Psi_w - \frac{r_w}{K_w}(F + W) \right] W - \delta_w W. \quad (5b)$$

and satisfy the principle of competitive exclusion in the sense that

$$\frac{\partial f}{\partial W} < 0, \quad \frac{\partial g}{\partial F} < 0$$

when $F > K_0$ is sufficiently large⁴. According to this principle, two species (or subpopulations) competing for the same resources cannot coexist, and only one of them should ultimately persist in the locality. In mathematical terminology, system (3) exhibits bistability which is frequency-dependent. The following theorem explicitly addresses this issue.

Theorem 1. Under the conditions (4), dynamical system (3) has four steady states (see Figure 1) in the region of biological interest $\Omega = \mathbb{R}_+^2 \setminus \{(0, 0)\}$ ⁵, namely:

- one nodal repeller $(K_b, 0)$ where $K_b > 0$ indicates the minimum viable population size (MVPS) of uninfected mosquitoes;
- one saddle point (F_c, W_c) of unstable coexistence of both mosquito species;
- two nodal attractors $(0, K_w)$ and $(K_*, 0)$, where $K_* > K_f$ defines the carrying capacity of wild females; only one of these steady states can be reached when $t \rightarrow \infty$ according to the initial conditions $F(0) > 0, W(0) > 0$ assigned to the system (3), namely:
 - If $F(0) > K_b$ and $W(0) > 0$ then $(K_*, 0)$ is reachable when $t \rightarrow \infty$ and wild mosquitoes should persist while *Wolbachia*-carriers become extinct (see Figure 2).

⁴ This condition is introduced in order to guarantee that uninfected female be able to find an uninfected partner for successful reproduction. Here we suppose parity between male and female progenies.

⁵ Here, we exclude the origin $(0, 0)$ since the total extinction of mosquito is unrealistic.

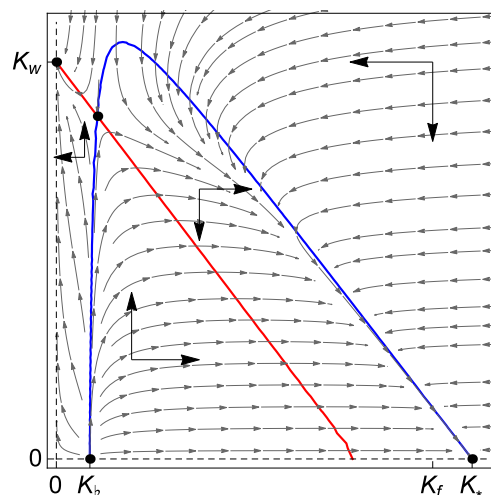


Fig. 1: Directional vector field and phase diagram of the dynamical system (3) in the plane (F, W) and its five steady state (black points); F -isocline is drawn in blue color and W -isocline is given in red color.

- If $F(0) < K_b$ and $W(0) > 0$ then $(0, K_w)$ is reachable when $t \rightarrow \infty$ and *Wolbachia*-carriers should persist while wild mosquitoes become extinct (see Figure 3).

Proof. To find the steady states of the dynamical system (3), we solve the system of algebraic equations

$$f(F, W) = 0, \quad g(F, W) = 0,$$

where $f(F, W)$ and $g(F, W)$ are defined by (5). As a result, we obtain

$$K_b = \frac{r_f K_0 + \Psi_f K_f - \sqrt{(r_f K_0 + \Psi_f K_f)^2 - 4r_f K_0 K_f (\Psi_f + \delta_f)}}{2r_f}, \quad (6)$$

$$K_* = \frac{r_f K_0 + \Psi_f K_f + \sqrt{(r_f K_0 + \Psi_f K_f)^2 - 4r_f K_0 K_f (\Psi_f + \delta_f)}}{2r_f}, \quad (7)$$

$$F_c = \frac{K_0 [\Psi_f (K_f - K_w) + \delta_f (K_f + K_w)]}{\Psi_f (K_f - K_w) + \delta_f K_w}, \quad W_c = K_w - F_c$$

To perform the stability analysis, we start with the most desirable steady state $(0, K_w)$ that implies the persistence of *Wolbachia*-carriers and extinction of wild mosquitoes (i.e., the population replacement). The Jacobian matrix $J(F, W)$ of the system (3) evaluated in this equilibrium

$$J(0, K_w) = \begin{pmatrix} \frac{K_w(\Psi_f - \delta_f) - K_f(\Psi_f + \delta_f)}{K_f} & 0 \\ -\Psi_w + \delta_w & -\Psi_w + \delta_w \end{pmatrix}$$

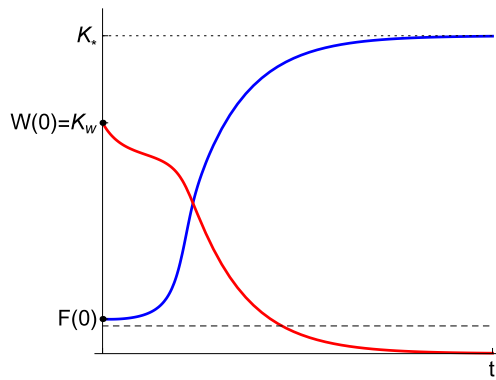


Fig. 2: Trajectories of the system (3) are attracted by $(K_*, 0)$ when $F(0) > K_b$, while $W(0) = K_w$; dashed line marks the value of K_b .

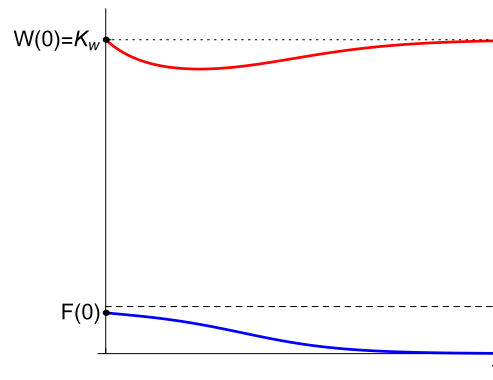


Fig. 3: Trajectories of the system (3) are attracted by $(0, K_w)$ when $F(0) < K_b$, while $W(0) = K_w$; dashed line marks the value of K_b .

is lower-triangular and its eigenvalues

$$\lambda_1 = (\Psi_f - \delta_f) \left[\frac{K_w}{K_f} - \frac{\Psi_f + \delta_f}{\Psi_f - \delta_f} \right] < 0,$$

$$\lambda_2 = -\Psi_w + \delta_w < 0$$

are both negative since $r_f = \Psi_f - \delta_f > 0$, $r_w = \Psi_w - \delta_w > 0$ and $K_w < K_f$. Therefore, $(0, K_w)$ is locally asymptotically stable (nodal attractor).

To analyze local stability of other three equilibria $(K_b, 0)$, (F_c, W_c) , and $(K_*, 0)$ the direct calculation of Jacobian eigenvalues results rather knotty and cumbersome from the computational standpoint. Alternatively, we propose to use Monte Carlo method (see, e.g. [17, 19]) to repeatedly verify the conditions $K_b > 0$, $F_c > 0$, $W_c > 0$, $K_* > 0$ and then calculate numerically the eigenvalues of the Jacobian evaluated at each steady state.

The coordinates of all three steady states can be expressed explicitly in terms of seven model's parameters $(\Psi_f, \Psi_w, \delta_f, \delta_w, K_f, K_0, K_w)$ whose baseline values are given in Table 1. The sampling pool $\mathbb{S} = \prod_{i=1}^7 P_i \in \mathbb{R}_+^7$ is defined by a Cartesian product of seven closed intervals of the form $P_i = [p_i - \theta p_i, p_i + \theta p_i]$ where each $p_i, i = 1, \dots, 7$ stands for the baseline value of one parameter (see Table 1) and $\theta > 0$ defines the variation range. Our sampling comprised 10^4 confounding scenarios $S = (s_1, \dots, s_7) \in \mathbb{S}$ where each $s_i \in P_i, i = 1, \dots, 7$ was randomly chosen for $\theta = 0.1$ (that is, with 10% deviation from the baseline values) under uniform distribution with no correlation between parameters. The result of numerical calculations are given in Figures 4, 5, and 6.

The upper chart on Figure 4 exhibits the coordinates of the coexistence steady state (F_c, W_c) in the phase plane (F, W) , two charts in the middle display the real parts of two eigenvalues (λ_1 and λ_2) of the Jacobian $J(F_c, W_c)$ versus the trial number $n = 1, \dots, 10^4$, while the lower

chart shows the distributions of the real parts of both eigenvalues.

It is worthwhile to note that each single point in the upper chart of Figure 4 is linked to one single point on the left chart and one single point on the right chart in the middle of Figure 4 and all these three points constitute together one segregated trial (or scenario $S = (s_1, \dots, s_7) \in \mathbb{S}$) where the values of seven parameters $(\Psi_f, \Psi_w, \delta_f, \delta_w, K_f, K_0, K_w)$ are randomly chosen from the sampling pool $\mathbb{S}$.

Since (F_c, W_c) has positive components, the coexistence of uninfected and *Wolbachia*-infected females is biologically viable, that is, $(F_c, W_c) \in \Omega$. However, $J(F_c, W_c)$ has one eigenvalue with positive real part ($\lambda_2 > 0$) and one eigenvalue with negative real part ($\lambda_1 < 0$). Therefore, the coexistence steady state is unstable (saddle point).

The upper and two central charts on Figure 5 exhibit the minimum viable population size K_b of wild mosquitoes and the real parts of eigenvalues λ_1 and λ_2 of the Jacobian $J(K_b, 0)$ versus the trial number $n = 1, \dots, 10^4$, respectively. The lower chart on Figure 5 shows the the distributions of the real parts of both eigenvalues. Since $K_b > 0$ in all trials, it is fair to affirm that $(K_b, 0) \in \Omega$. However, $J((K_b, 0))$ has two eigenvalues with strictly positive real parts and, therefore, the equilibrium $(K_b, 0)$ is unstable (nodal repeller). The latter is also illustrated by the behavior of system trajectories $F(t)$ and $W(t)$ that utterly depends on the choice of initial conditions (see Figures 2 and 3).

Finally, the upper and two central charts on Figure 6 exhibit the carrying capacity K_* of wild mosquitoes and the real parts of eigenvalues λ_1 and λ_2 of the Jacobian $J(K_*, 0)$ versus the trial number $n = 1, \dots, 10^4$, respectively. The lower chart on Figure 6 presents the distributions of the real parts of both eigenvalues. It is easy to see that $K_* > K_w > 0$ in all trials meaning that $(K_*, 0) \in \Omega$. Additionally, $J((K_*, 0))$ has two eigenvalues

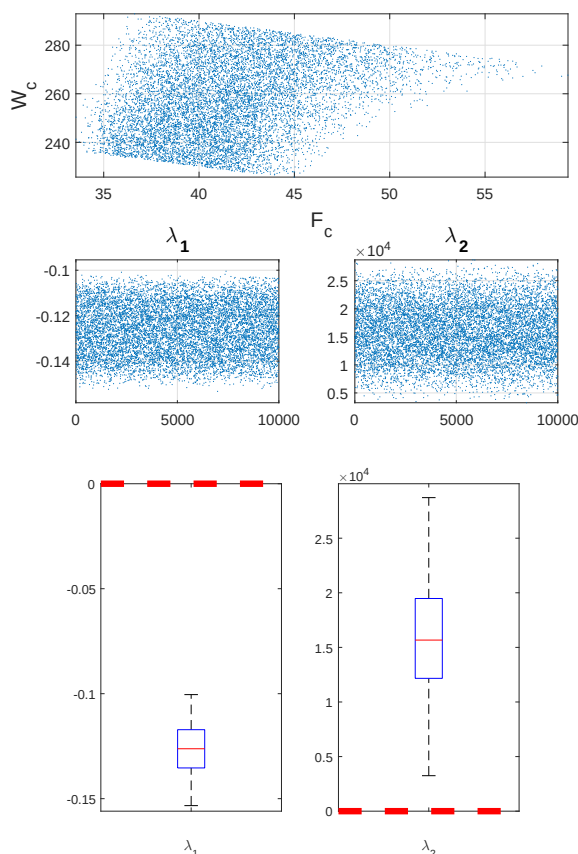


Fig. 4: Coexistence equilibrium (F_c, W_c) in the phase plane (F, W) (top); eigenvalues λ_1 and λ_2 of $J(F_c, W_c)$ versus trial number $n = 1, \dots, 10^4$ (middle); distribution of both eigenvalues λ_1 and λ_2 (bottom).

with strictly negative real parts and, therefore, the equilibrium $(K_*, 0)$ is locally asymptotically stable (nodal attractor). This completes the proof of the Theorem 1. $\square$

Remark 2 It is worthwhile to note that the competition system (3) exhibits the phenomenon of so-called stiffness, which can be detected by using the definition of stiffness ratio (see, e.g. [18] or similar textbooks). In mathematical terms, the stiffness ratio ρ is the quotient between the absolute values of maximal and minimal eigenvalues (in terms of their real parts), that is,

$$\rho = \frac{\max\{|\Re(\lambda_1)|, |\Re(\lambda_2)|\}}{\min\{|\Re(\lambda_1)|, |\Re(\lambda_2)|\}}.$$

When ρ is very large, the dynamical system is said to be stiff and it is exactly what we have observed in the proof of the Theorem 1.

Theorem 1 clearly states that coexistence of two mosquito species is hardly expected and that only one subpopulation can ultimately survive and persist in the

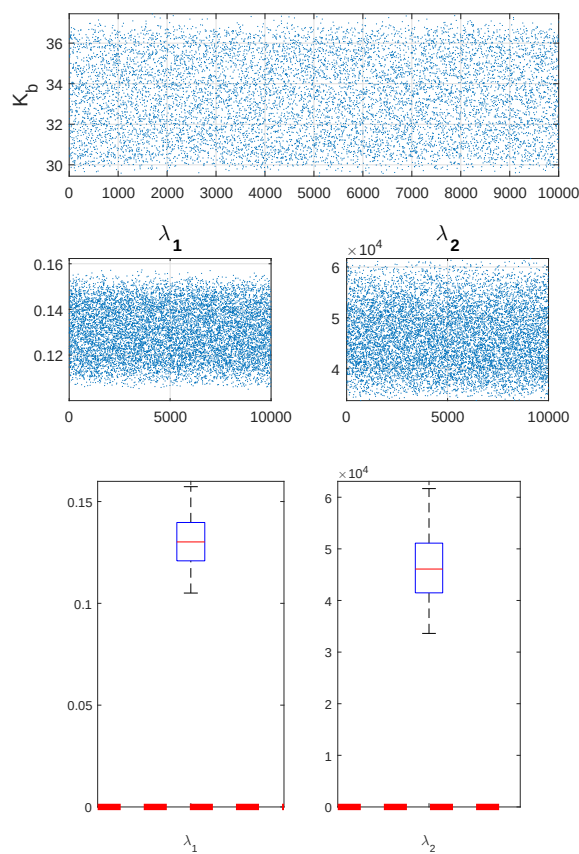


Fig. 5: Minimum viable population size K_b of uninfected mosquitoes versus the trial number $n = 1, \dots, 10^4$ (top); eigenvalues λ_1 and λ_2 of $J(K_b, 0)$ versus the trial number $n = 1, \dots, 10^4$ (middle); distribution of both eigenvalues λ_1 and λ_2 (bottom).

locality. The final outcome of the competition depends entirely on the choice of initial conditions, and the latter agrees with bistable nature of the competition system (3). Figure 1 shows that the attraction basin of *Wolbachia*-persistence steady state $(0, K_w)$ is rather small in comparison to the attraction basin of *Wolbachia*-free steady state $(K_*, 0)$.

In practical terms, however, we know that only wild mosquitoes $F(t)$ are initially presents in the target locality and their density should be around equilibrium (i.e., carrying capacity). Figure 1 also shows that when $K_f < F(0) < K_*$ the *Wolbachia*-free steady state $(K_*, 0)$ will be reachable no matter how high is $W(0)$. Therefore, a single abundant release of *Wolbachia*-carriers will not establish *Wolbachia* infection among wild mosquito population and this argument is supported by practical experience [28, 42].

As an alternative, we propose to apply the optimal control approach in order to derive the control actions based on multiple periodic releases of *Wolbachia*-infected females.

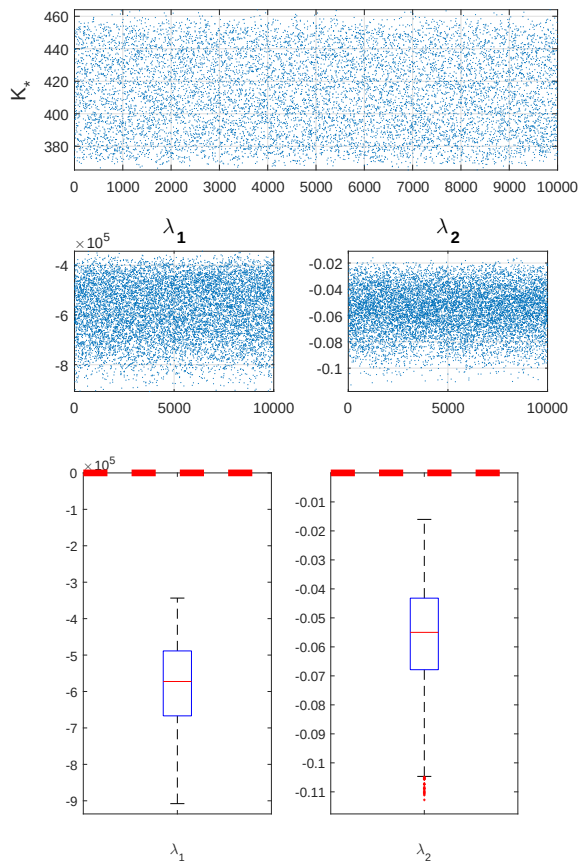


Fig. 6: Carrying capacity K_* of wild mosquitoes versus the trial number $n = 1, \dots, 10^4$ (top); eigenvalues λ_1 and λ_2 of $J(K_*, 0)$ versus the trial number $n = 1, \dots, 10^4$ (middle); distribution of the eigenvalues λ_1 and λ_2 (bottom).

3 Optimal control approach

The optimal control approach requires to define the control variable that mathematically expresses the control action. In our case, let $u(t) : [0, T] \mapsto [0, u_{\max}]$ be the number of *Wolbachia*-carriers to be released at the day t in the target locality. Here, $u_{\max} > 0$ stands for the maximum number of *Wolbachia*-infected females available for a daily release, and this number naturally depends on the production capacity of a laboratory where *Wolbachia*-carriers are cultivated. We also suppose that the overall time $T > 0$ of control action is finite but is set free.

Wolbachia-based biocontrol consists in replacing wild mosquitoes (fully capable of transmitting infections) by *Wolbachia*-infected ones (whose transmission capacity is significantly reduced) and there are two important criteria to be taken into account for design of optimal release strategies:

1. To reach the population replacement as soon as possible since ongoing releases are associated with

some costs (maintenance of the laboratory, personnel, etc.).

2. To release as little mosquitoes as possible since their artificial breeding also come at considerable costs.

In mathematical terms, the purpose of control action is to find an optimal release program $u^*(t) \in [0, u_{\max}]$, $t \in [0, T^*]$ and a minimum time $T^* \in (0, \infty)$ satisfying the endpoint condition

$$F(T^*) = F_T < K_b \quad (8)$$

while minimizing the objective functional

$$\min_{\substack{0 \leq u(t) \leq u_{\max} \\ 0 < T < \infty}} \mathcal{J}(u, T) = \min_{\substack{0 \leq u(t) \leq u_{\max} \\ 0 < T < \infty}} \int_0^T \left(1 + \frac{C}{2} u^2(t) \right) dt \quad (9)$$

over the set of all possible solutions of the dynamical system

$$\frac{dF}{dt} = \left[\Psi_f - \frac{r_f}{K_f} (F + W) \right] F \left(\frac{F}{K_0} - 1 \right) - \delta_f F, \quad (10a)$$

$$\frac{dW}{dt} = \left[\Psi_w - \frac{r_w}{K_w} (F + W) \right] W - \delta_w W + u(t). \quad (10b)$$

with initial conditions

$$F(0) = F_0, \quad W(0) = 0. \quad (11)$$

The endpoint condition (8) implies that, under optimal release program $u^*(t)$, the population of wild females should drop below the minimum viable population size K_b by the end of control intervention period $[0, T^*]$. The latter should guarantee the eventual population replacement.

The objective functional (9) is defined in accordance with two above mentioned criteria. Namely, it refers to minimization of the terminal time, $\int_0^T dt$, together with the cumulative control effort $\int_0^T \frac{C}{2} u^2(t) dt$ (i. e., cultivation costs of *Wolbachia*-infected mosquitoes in laboratory conditions) over the period $[0, T]$. In (9), $C > 0$ expresses the (relative) unit cost of control action (i.e., production of one cohort of *Wolbachia*-infected females). By varying the value of C , one can reflect different priorities of the decision-making. Smaller values of C in the objective functional (9) would imply that production costs are (relatively) low in comparison to the time appreciation by the decision-maker, while higher values of C would imply that production costs are (relatively) high.

In formulation of the control problem we have assumed no linear relationship between the coverage of control actions and their underlying costs, while the cost related to the intervention period, T , has been supposed proportional to time. Under such assumptions, the integrand function in (9) is quadratic with respect to control variable. This approach is very common in dynamic optimization that involves population dynamics,

and its justifications can be consulted in [2, 27, 29, 30, 36]. Moreover, the quadratic form of control in (9) is helpful for justifying the existence of solution of the optimal control problem (8)–(11) and provides very logical and simple interpretation of the maximum principle (see Remark 3).

The problem of minimizing the objective functional (9) subject to dynamical constraints (10) with initial conditions (11) and terminal condition (8) can be solved by applying the Pontryagin maximum principle. The convexity of the integrand in (9) with respect to control variable u , the linearity of the ODE system (10) in u , and the compactness of the range of state variables in $\Omega \subset \mathbb{R}_+^2$ for any finite $0 < T < \infty$ should altogether assure the existence of the optimal control for sufficiently large $u_{\max}$ (see more details and formal proofs in the book by [12]).

In the context of our problem, the variant of maximum principle dealing with free terminal time [21] will be particularly useful. According to this variant, an optimal pair (u^*, T^*) must always satisfy the necessary conditions that are formulated in terms of so-called Hamiltonian function:

$$H(F, W, u, \lambda_1, \lambda_2) = -1 - \frac{C}{2}u^2 \quad (12)$$

$$+ \lambda_1 \cdot \left\{ \left[\Psi_f - \frac{r_f}{K_f}(F+W) \right] F \left(\frac{F}{K_0} - 1 \right) - \delta_f F \right\}$$

$$+ \lambda_2 \cdot \left\{ \left[\Psi_w - \frac{r_w}{K_w}(F+W) \right] W - \delta_w W + u(t) \right\}$$

where λ_1, λ_2 can be viewed as Lagrange multipliers.

Let (u^*, T^*) be an optimal pair in the sense that $u^*(t)$ is a piecewise continuous real function with domain $[0, T^*]$ and range $[0, u_{\max}]$ and $\mathcal{J}(u^*, T^*) \leq \mathcal{J}(u, T)$ for all other controls u and times T . Let $F^*(t) = F(t, u^*(t))$ and $W^*(t) = W(t, u^*(t))$ be the corresponding states defined for all $t \in [0, T^*]$. Then there exist two piecewise differentiable adjoint functions $\lambda_1 : [0, T^*] \mapsto \mathbb{R}, i = 1, 2$ satisfying the adjoint system of differential equations

$$\frac{d\lambda_1}{dt} = - \frac{\partial H(F^*, W^*, u^*, \lambda_1, \lambda_2)}{\partial F}, \quad (13a)$$

$$\frac{d\lambda_2}{dt} = - \frac{\partial H(F^*, W^*, u^*, \lambda_1, \lambda_2)}{\partial W} \quad (13b)$$

with transversality condition

$$\lambda_2(T^*) = 0 \quad (14)$$

while $0 < T^* < \infty$ satisfies the condition (8) and fulfills that

$$H(F^*(T^*), W^*(T^*), u^*(T^*), \lambda_1(T^*), \lambda_2(T^*)) = 0. \quad (15)$$

Moreover, the Hamiltonian (12) has a critical point (maximum⁶) at $u = u^*(t)$, that is,

$$H(F^*(t), W^*(t), u^*(t), \lambda_1(t), \lambda_2(t))$$

⁶ It is easy to verify that $\frac{\partial^2 H}{\partial u^2} = -C < 0$ for all admissible u .

$$\geq H(F^*(t), W^*(t), u(t), \lambda_1(t), \lambda_2(t))$$

for any admissible $u(t) : [0, T^*] \mapsto [0, u_{\max}]$ and for almost all $t \in [0, T^*]$.

According [21], the above condition can be written in a more convenient way, namely:

$$\left. \begin{aligned} u^*(t) &= 0 & \text{if } \frac{\partial H}{\partial u} < 0 \\ 0 < u^*(t) < u_{\max} & \text{if } \frac{\partial H}{\partial u} = 0 \\ u^*(t) &= u_{\max} & \text{if } \frac{\partial H}{\partial u} > 0 \end{aligned} \right\} \quad (16)$$

or, equivalently,

$$u^*(t) = \max \left\{ 0, \min \left\{ \frac{\lambda_2}{C}, u_{\max} \right\} \right\}. \quad (17)$$

Remark 3 The Pontryagin maximum principle gives us some interesting insights regarding the costs of control strategies. From the economics standpoint, the condition

$$\frac{\partial H}{\partial u} = -Cu + \lambda_2 = 0$$

implies that, under optimal release rate u^* , the marginal cost of control action (expressed by the term $Cu(t)$, i.e. the number of released Wolbachia-carriers) should be equal to its marginal benefit (given by the term λ_2). If the marginal cost of u^* is higher than its marginal benefit (that is, $\frac{\partial H}{\partial u} < 0$ in (16)) then it is optimal not to employ this strategy at all, i.e., $u^*(t) = 0$. Alternatively, if the marginal cost of u^* is lower than its marginal benefit (that is, $\frac{\partial H}{\partial u} > 0$ in (16)) then it is optimal to use all available resources, i.e., $u^*(t) = u_{\max}$.

Using the compact form of $u^*(t)$ from (17) (which is also known as the *characterization* of optimal control) we can reduce the original optimal control problem (8)–(11) to a two-point boundary value problem which is usually called the *optimality system*. The latter is conformed by four differential equations with four endpoint conditions, namely:

- two *direct* equations (10) where $u(t)$ is replaced by its characterization (17);
- two *adjoint* equations (13) where $u(t)$ is replaced by its characterization (17);
- two initial conditions (11) specified at $t = 0$;
- one transversality condition (14) and one endpoint condition (8) specified at $t = T^*$.

The optimal time $0 < T^* < \infty$ is then determined from the optimality condition (15).

The optimality system described above does not have a closed-form solution and can only be solved

numerically. Traditional techniques for solving the optimality systems are based on so-called *forward-backward sweep methods* [21], *shooting methods* [33], and *direct collocation methods* [1]. However, these techniques do not guarantee the convergence of the numerical algorithm when the final time T is left free or when additional constraints of the type (8) are imposed.

On the other hand, the next-generation optimal software package GPOPS-II solver⁷ designed for MATLAB platform [31] is capable of dealing with time-optimal control problems since it does not rely on numerical integration and implements an adaptive combination of two collocation techniques (direct and orthogonal collocation) known as *Radau pseudospectral method* (see more details in [13]). Due to its adaptiveness, the GPOPS-II algorithm is applicable for routine solutions of stiff systems (see Remark 2) and, therefore, it is the most suitable software tool for solving the control problem (8)-(11).

4 Numerical results and discussion

Here we propose to deal with wMelPop strain of *Wolbachia* which is considered the best one for prevention and control of arboviral infections [10,11,26,40]. However, many scholars point out that wMelPop strain of *Wolbachia* is difficult to establish in wild *Aedes aegypti* populations due to its high fitness cost [15,24,25,32,34,35], what makes our task of solving the optimal control problem (8)-(11) even more challenging.

Table 1 assigns the numerical values to all constant parameters appearing in the competition system (3), (10) in accordance with scientific evidence available in literature with regards to wMelPop *Wolbachia* strain. Here, mosquitoes' birth and death rates ($\Psi_f, \Psi_w, \delta_f, \delta_w$) are measured in day^{-1} while (K_f, K_0, K_w) are expressed in terms of the numbers of individuals (e.g., hundreds, thousands, etc.). It is worthwhile to note that for parameter values from Table 1, we have

$$K_* = 413.168, \quad F_c = 40.943, \\ K_b = 33.327, \quad W_c = 259.057.$$

Remark 4 It is rather difficult to estimate the size of mosquito population in a particular locality. However, it has been estimated [36,37] that, in dengue-endemic areas, there are usually between 1 and 2 *Aedes aegypti* females per one human host. Therefore, the parameter values in Table 1 qualitatively represent the mosquito population (in thousands of individuals) in a medium-sized city with population of about 200.000-400.000 inhabitants.

We suppose that at $t = 0$ there is no *Wolbachia*-carriers in the target locality, that is, $W(0) = 0$

⁷ For more information regarding GPOPS-II solver please visit <http://gpops2.com/>

Table 1: Parameter values for the competition model (3), (10).

Parameter	Range	References
$\Psi_f = 0.32667$	0.28 - 0.38	[6,8,38]
$\Psi_w = 0.21333$	0.18 - 0.25	[15,24,25,35]
$\delta_f = 0.03333$	1/8 - 1/42	[6,8,38]
$\delta_w = 0.06666$	2/8 - 2/42	[15,24,25,35]
$K_f = 374$		assumed
$K_w = 300$		assumed
$K_0 = 30$		assumed

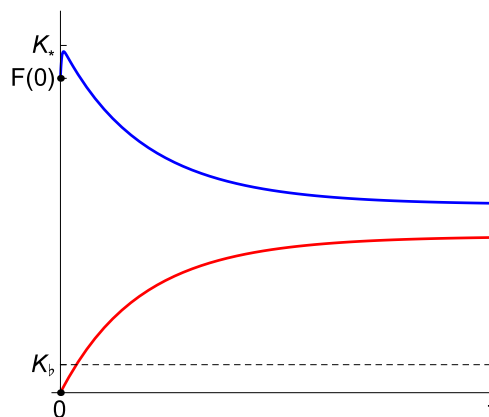


Fig. 7: Trajectories of the system (10) under constant control $u(t) = u_{\max}$ when $u_{\max}$ is not high enough; $F(t)$ and $W(t)$ are drawn by blue and red curves, respectively, and the dashed line marks the minimum viable population size K_b .

and assign the following initial and endpoint conditions to wild *Aedes aegypti* females:

$$F(0) = K_f < K_*, \quad F(T) = K_0 < K_b. \quad (18)$$

The above initial condition is more realistic than $F(0) = K_*$ due to the vector control measures (such as insecticide/larvicide spraying, lethal ovitraps, etc.) that are continuously performed in the areas dominated by wild *Aedes aegypti* mosquitoes. Additionally, the value $u_{\max}$ (i.e., the number of *Wolbachia*-carrying females to be release daily) should be sufficiently high in order to avoid situations presented in Figure 7 and to guarantee the existence of solution of the optimal control problem (8)-(11). Effectively, if $u_{\max}$ is not high enough, the *Wolbachia*-carriers are unable to reach the frequencies sufficient for their proliferation (even when $u(t) = u_{\max}$ and $t \rightarrow \infty$) due to their reduced fitness.

As we have mentioned in Section 3, the choice of the coefficient $C > 0$ defines the priorities of the decision-making. Therefore, it is reasonable to consider two alternative priorities in decision-making, namely:

Option A. Time is far more important than the production cost of *Wolbachia*-infected mosquitoes ($C = 0.01$);

Option B. Both time and the production cost are equally important ($C = 1$);

On the other hand, one may intuitively perceive the existence of a certain tradeoff between the maximal capacity of daily releases, $u_{\max}$, and the expected length of control intervention period, T . In order to illustrate such tradeoff, let us consider four different cases defined as

Case 1: $u_{\max} = 15$; **Case 2:** $u_{\max} = 20$;
Case 3: $u_{\max} = 25$; **Case 4:** $u_{\max} = 30$.

Thus, we can set up eight different scenarios by coupling one option (**Options A and B**) and one case (**Cases 1, 2, 3, and 4**). Under this setting, **Scenario 1A** combines **Option A** with **Case 1**.

All eight scenarios have been simulated under the same values of all constant parameters (see Table 1) and with the same initial and terminal conditions (18). We have also assumed the standard GPOPS-II numerical tolerance of 10^{-5} for internal calculations with regards to *scaling*⁸. It is worthwhile to note that GPOPS-II package does not deliver solution to a free- or minimum-time optimal control problem unless the optimality condition (15) is satisfied and in all subsequent numerical experiments GPOPS-II has delivered the underlying solutions.

Let us consider first **Option A** where the promptness of intervention is prioritized over the costs related to artificial breeding of *Wolbachia*-carriers. The results of numerical solutions of the optimal control problem (8)-(11) for **Scenarios 1A-4A** are displayed in Figures 8 and 9.

When the maximum daily release capacity $u_{\max}$ is low (cf. upper rows in Figures 8 and 9 where $u_{\max} = 15$), the optimal time for policy implementation is rather long ($T^* = 288$ days) despite the priority being put on time. The release strategy in this case (**Scenario 1A**, upper chart in Figure 8) consists in two actions:

1. During the first 266 days, the daily releases of *Wolbachia*-carrying females are performed at the maximum capacity $u_{\max}$;
2. For the last 22 days of intervention, the daily release capacity is sharply reduced from $u_{\max}$ to 0.

The effect of this intervention policy over the mosquito population is shown in the upper chart of Figure 9. It is worthwhile to note that the population of wild mosquitoes $F^*(t)$ (blue curve) decreases during the whole intervention period $[0, T^*]$, crosses the horizontal line $F(t) = K_0$ of minimum viable population size, and hits the value $K_0 < K_1$ exactly at $T^* = 288$. However, the trajectory of $F^*(t)$ has an inflexion point around $t = 150$. The latter implies that the rate of decrease is higher in the

beginning of the intervention than in the end. Therefore, during the period $[150, T^*]$ not all released *Wolbachia*-carriers effectively contribute to the population replacement, and at least some part of them is merely lost to the competition with coevals (infected and uninfected).

The above-mentioned shortcoming can be eliminated by increasing the maximum daily release capacity to $u_{\max} = 20$. As displayed in the second row of Figure 9 (**Scenario 2A**), the trajectory of $F^*(t)$ has no inflexion and the overall period of control intervention is reduced to $T^* = 91$ days (that is, by almost 70%). Additionally, the release strategy in this case (**Scenario 2A**, second-row chart in Figure 8) now consists in three actions:

1. At the outset of 8 days, the daily releases are sternly increased from 14 to $u_{\max} = 20$;
2. During the consequent 58 days (i.e., for $t \in [9, 66]$), the daily releases are performed at the maximum capacity $u_{\max}$;
3. For the last 25 days of intervention ($t \in [67, 91]$), the capacity of daily release is sternly reduced from $u_{\max}$ to 0.

The effect of this intervention policy over the mosquito population is shown in the second-row chart of Figure 9.

Similar tendencies are observed in the forms of the optimal release policies $u^*(t)$ and the underlying states $F^*(t), W^*(t)$ when the maximum daily release capacity is raised further to $u_{\max} = 25$ and $u_{\max} = 30$ what corresponds to **Scenarios 3A** and **4A** (see the charts on the third and fourth rows of Figures 8 and 9, respectively). However, the time lapses of performing the releases of maximum daily capacities $u_{\max}$ are gradually becoming shorter as such capacity $u_{\max}$ increases.

Quite expectedly, the overall period of control intervention is also reduced from $T^* = 91$ to $T^* = 69$ days (or by 24% – **Scenario 2A** versus **Scenario 3A**) and from $T^* = 69$ to $T^* = 62$ days (or by 10% – **Scenario 3A** versus **Scenario 4A**). Therefore, further enhancement of $u_{\max}$ looks unnecessary, since the expected gain in time becomes smaller and smaller.

Now let us consider **Option B** where both the promptness of intervention and the costs related to artificial breeding of *Wolbachia*-carriers have the same priority. The results of numerical solutions of the optimal control problem (8)-(11) for **Scenarios 1B-4B** are displayed in Figures 10 and 11.

Effectively, the optimal release policies, $u^*(t)$ designed for **Scenarios 1B-3B** (see three upper rows in Figure 10) bear strong resemblance with optimal control profiles obtained for **Scenarios 2A-4A** (cf. last three rows in Figure 8) since all these policies consist of three control actions. Namely, at the outset of control intervention, the intensity of releases is progressively increased for about 45-55 days from a very low level $u^*(0) = 2$ towards $u_{\max}$.

⁸ GPOPS-II scales automatically all input intervals $[0, T]$ to the interval $[-1, 1]$ (see more detailed information in [31].)

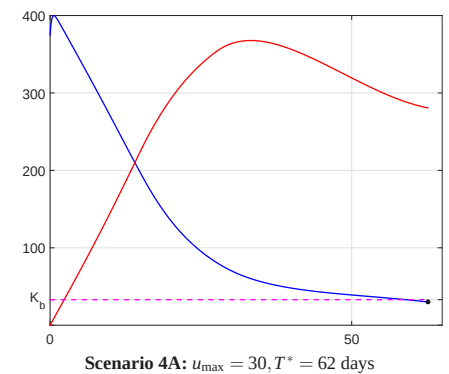
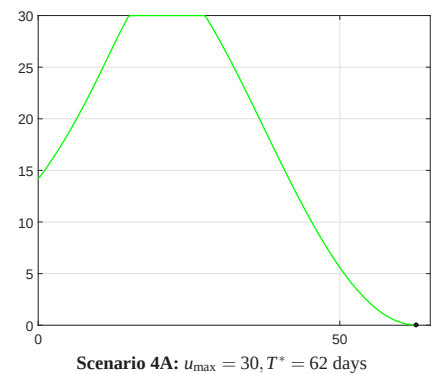
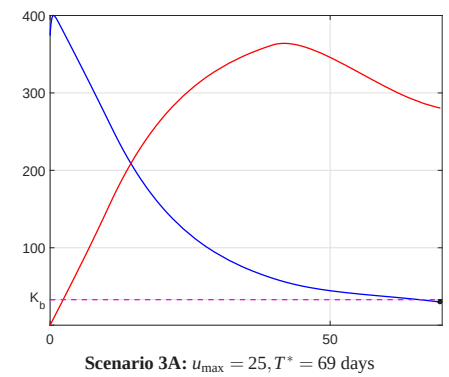
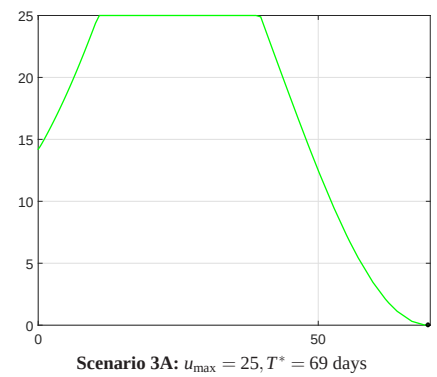
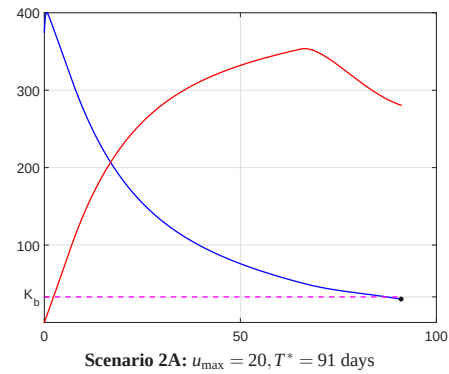
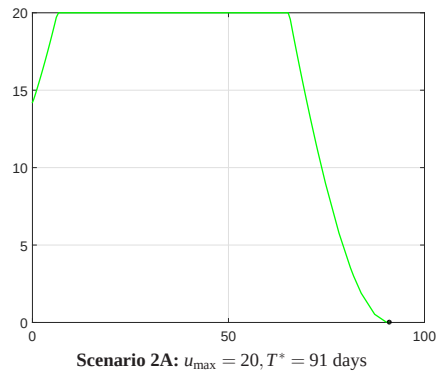
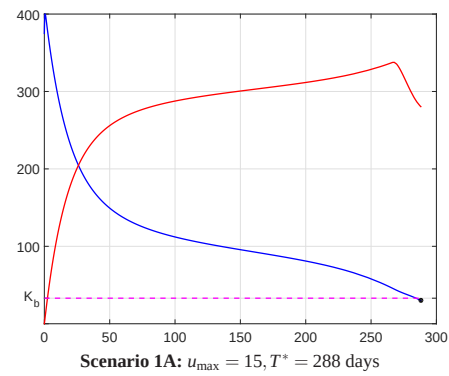
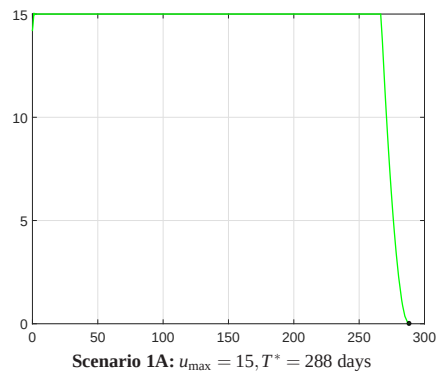


Fig. 8: Optimal control $u^*(t)$ for **Scenarios 1A-4A**.

Fig. 9: Optimal trajectories $F^*(t)$ (blue) and $W^*(t)$ of the system (10) under optimal control $u^*(t)$ for **Scenarios 1A-4A**.

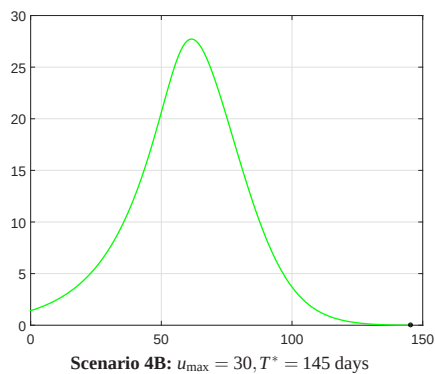
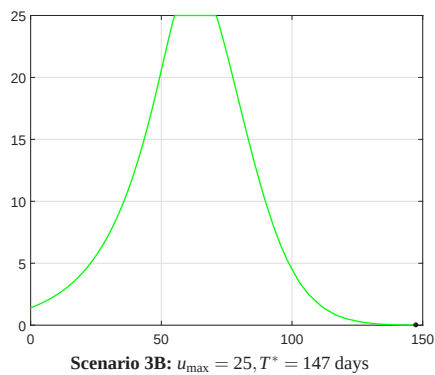
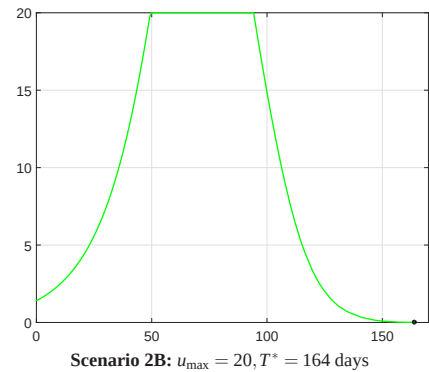
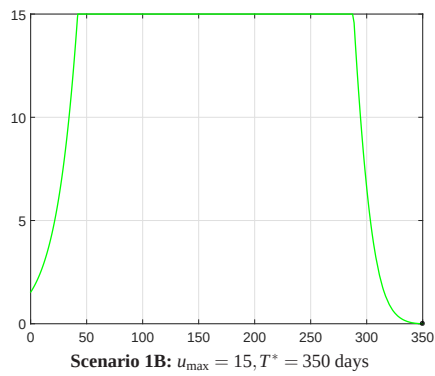


Fig. 10: Optimal control $u^*(t)$ for Scenarios 1B-4B.

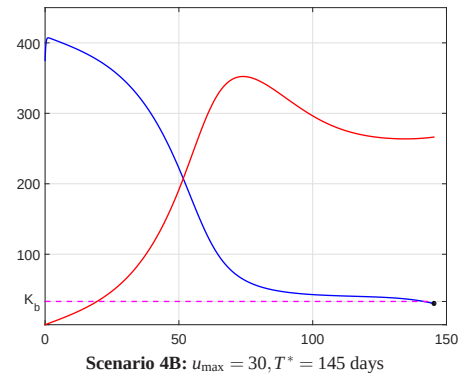
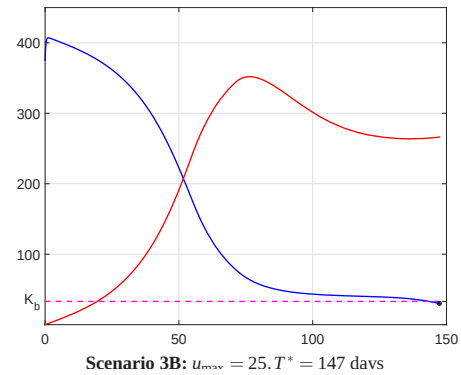
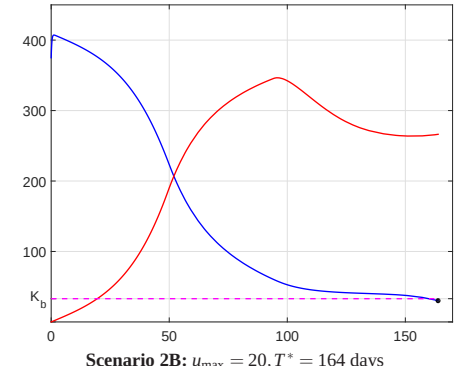
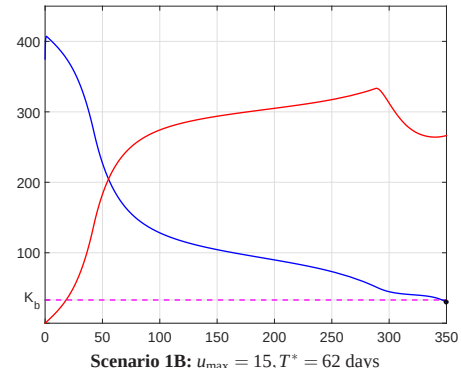


Fig. 11: Optimal trajectories $F^*(t)$ (blue) and $W^*(t)$ of the system (10) under optimal control $u^*(t)$ for Scenarios 1B-4B.

Further, daily releases are performed at the level of maximum release capacity $u_{\max}$ for a certain period of time, which becomes shorter for higher values of $u_{\max}$. Finally, the intensity of daily releases descends gradually from $u_{\max}$ towards 0.

On the other hand, the optimal release policy designed for **Scenario 4B** (bottom row in Figure 10) is totally smooth (while all other optimal controls obtained in this study are truncated) and this particular control policy does not require to perform releases at the maximum capacity $u_{\max}$.

As expected, the overall time of control intervention is longer for lower values of $u_{\max}$. Thus, for $u_{\max} = 15$ (**Scenario 1B**) we have $T^* = 350$ day and an essential gain in time of about 53% is obtained by raising the maximum daily release capacity to $u_{\max} = 20$ (cf. **Scenario 2B** with $T^* = 164$). However, further increase in $u_{\max}$ shows lesser effect on the reduction in time. In particular, the overall period of control intervention is reduced from $T^* = 164$ to $T^* = 147$ days (or by 10% – **Scenario 2B** versus **Scenario 3B**) and from $T^* = 147$ to $T^* = 145$ days (or by 1% only – **Scenario 3B** versus **Scenario 4B**).

It is worthwhile to note that, in all four **Scenarios 1B-4B**, the population trajectory of wild mosquitoes $F^*(t)$ (blue curves in Figure 11) decreases under the corresponding $u^*(t)$, crosses the threshold of minimum viable population size K_0 , and arrives to the endpoint condition $F(T) = K_0$ exactly at $T = T^*$. However, in all four **Scenarios 1B-4B** the trajectories of $F^*(t)$ are not strictly convex and exhibit a “concave-convex” pattern within $[0, T^*]$. In other words, during the “concave” lags not all released *Wolbachia*-carriers effectively contribute to the population replacement, and at least some part of them is merely lost to the competition with coevals (infected and uninfected).

From the foregoing analysis, we can conclude that the release policies obtained under **Option B (Scenarios 1B-4B)** are less efficient than those designed under **Option A (Scenarios 1A-4A)**. To support this argument, let us now estimate the total (cumulative) number of *Wolbachia*-carrying females (NWC) that are needed for implementation of the optimal release policy $u^*(t)$:

$$\text{NWC}(u^*) = \int_0^{T^*} u^*(t) dt.$$

Table 2 summarizes the results for all eight scenarios and clearly points out that **Scenario 4A** renders the best outcome in terms of minimal time ($T^* = 62$ days) and minimal number of *Wolbachia*-carries (1.1195×10^3) needed for the population replacement.

On the other hand, the form of optimal control function $u^*(t), t \in [0, 62]$ obtained for **Scenario 4A** (see Figure 8, bottom chart) only gives the general idea for practical implementation of the underlying release program. Performing continuous releases is a challenging task from the practical point of view. Therefore, this

Table 2: A synoptic outline of eight scenarios

Scenario	$u_{\max}$	T^*	$\text{NWC}(u^*)$
1A	15	288	4.1120×10^3
1B	15	350	4.1355×10^3
2A	20	91	1.4731×10^3
2B	20	164	1.5460×10^3
3A	25	69	1.2078×10^3
3B	25	147	1.3274×10^3
4A	30	62	1.1195×10^3
4B	30	145	1.2987×10^3

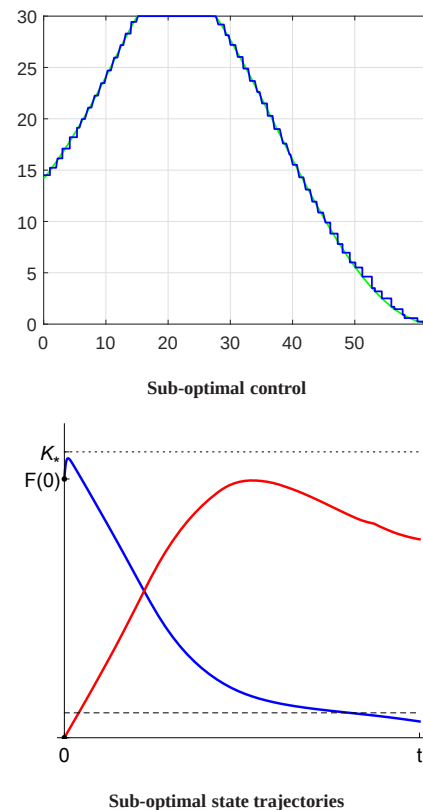


Fig. 12: Sub-optimal control strategy $\hat{u}^*(t)$ (upper chart) and corresponding trajectories $\hat{F}^*(t)$ (blue) and $\hat{W}^*(t)$ (red) of the system (10) for **Scenario 4A**.

optimal control can be approximated by a *sub-optimal* piecewise constant control $\hat{u}^*(t)$, defined in the following way:

$$\hat{u}^*(t) = \int_{i-1}^i u^*(t) dt \text{ for } t \in [i-1, i) \text{ and } i = 1, 2, \dots, 62.$$

The latter implies that during each i -th day, the number of released mosquitoes is equal to the average value of $u^*(t)$

taken over this day, that is, for $t \in [i-1, i]$. Figure 12 provides the graphs for $\hat{u}^*(t)$ and corresponding state trajectories. It is worthwhile to point out that the uprise in total (cumulative) number of *Wolbachia*-carrying females (NWC) to be released under sub-optimal strategy $\hat{u}^*(t)$ is negligible (less than 0.06%) while its outcome is favorable in the sense of driving the population of $F(t)$ towards extinction. Note that sub-optimal trajectories $\hat{F}^*(t)$ and $\hat{W}^*(t)$ in the lower chart in Figure 12 displays simulations for $t \in [0, 78]$, that is, 15 days more after suspension of the releases (i.e., with $\hat{u}^*(t) = 0$ for $t \in [62, 78]$).

To complete this section, let us accentuate that, in all eight scenarios, the optimal control problem (8)-(11) has exhibited an anticipated level of sensitivity with respect to slight changes in terminal (8) and initial (11) conditions. By performing a series of numerical experiment with slightly perturbed values of $F(0)$ and $F(T^*)$, we have detected some consistent patterns which we should briefly describe further on without presenting here the corresponding graphics of $u^*(t)$ and $F^*(t), W^*(t)$.

When $F(0)$ in (11) is altered upwards (that is, $K_f < F(0) < K_*$), it extends the overall time T^* of the optimal control policy $u^*(t)$ together with a length of the control action where $u^*(t) = u_{\max}$. On the other hand, when $F(0)$ is altered downwards (that is, $K_b < F(0) < K_f$), both T^* and the action with $u^*(t) = u_{\max}$ get shorten down.

An opposite pattern has been observed regarding the variation of the terminal condition (8). When $F(T^*)$ is altered downwards (that is, $0 < F(T^*) < K_0$), the overall time T^* and the control action with $u^*(t) = u_{\max}$ are both extended, while an upward alteration of $F(T^*)$ (that is, $K_0 < F(T^*) < K_b$) results in the shorter T^* and a shorter action with $u^*(t) = u_{\max}$.

5 Conclusions

In this paper, we have presented a stylized population dynamics model focused on the competition between two subpopulations of *Aedes aegypti* females sharing the same locality and underlying resources (food, breeding sites, etc.). Both subpopulations (*Wolbachia*-free and *Wolbachia*-carrying females) possess different fitness characteristics (such as fecundity and longevity) while their reproduction success depend on the frequency of *Wolbachia* infection. Namely, *Wolbachia*-infected females, despite their lower individual fitness, have more chances for producing viable offspring than their uninfected coevals due to the phenotype of cytoplasmic incompatibility induced by *Wolbachia* infection. Using this population dynamics model, we have shown that the coexistence of two mosquito subpopulations is unstable and the final outcome of the competition essentially depends on the frequency of *Wolbachia* infection. The latter also agrees with the principle of competitive exclusion and corroborates the results obtained by other scholars [3, 5, 6, 35, 39].

Our principal goal was to design a control intervention policy that relies on daily releases of *Wolbachia*-infected females, facilitates the *Wolbachia* invasion among wild population of *Aedes aegypti* mosquitoes, and ultimately ensures the population replacement by driving the population of wild mosquitoes towards extinction. The latter has been accomplished by applying the modeling framework of optimal control and introducing the optimization criterion (9) that seeks out to minimize the overall time of the policy implementation together with its underlying costs while ensuring the fulfilment of the endpoint condition (8).

By considering eight different scenarios engendered by two priorities in decision-making (**Option A** and **Option B**, Section 4) together with variation of the daily release capacities (**Cases 1-4**, Section 4) we have shown that *wMelPop Wolbachia* strain should be capable of invading the population of wild mosquitoes despite its negative effect on the mosquito individual fitness. However, the control intervention programs targeting to establish this particular strain are quite atypical and bear no resemblance to the release policies traditionally employed by practitioners in field releases.

In this paper, we have not considered the effect of *wMelPop Wolbachia* strain on transmission of dengue and other vector-borne diseases, and this is left for future research.

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